

TaxNexus Production Deploy

Runbook

Submission runbook — SvelteKit + Strapi + nginx/TLS + MySQL + Mautic on Hostinger KVM2

Project	TaxNexus (soloai_starter)
App domain	taxnexusapp.com
CMS domain	cms.taxnexusapp.com
Public IPv4	2.25.203.69
Host	Hostinger KVM2 · Docker Compose
Verified	2026-08-14 · LIVE 200 OK

1. VPS Details

Item	Value
Public IPv4	2.25.203.69
Provider	Hostinger KVM2 (Docker Engine + Compose v2)
App subdomain	taxnexusapp.com → SvelteKit app (adapter-node, :3000)
CMS subdomain	cms.taxnexusapp.com → Strapi admin/API (:1337)
www alias	www.taxnexusapp.com → app (in cert SAN)
Edge / TLS	nginx on :80/:443, one Let's Encrypt cert covering all three names

The app reaches Strapi **server-side over HTTPS** at `https://cms.taxnexusapp.com` (hairpins back through nginx), so **cms must be in the TLS certificate** — confirmed in the SAN in §4.

2. Full Contents — docker-compose.yml

Verbatim from repo root. Defines the six-part stack: db (MySQL), nginx, strapi, app (SvelteKit), certbot, and the mautic_* services.

docker-compose.yml

repo root

```
x-soloai-volumes:
  &soloai-volumes
  - ${BASE_PATH}/mautic/config:/var/www/html/config:z
  - ${BASE_PATH}/mautic/logs:/var/www/html/var/logs:z
  - ${BASE_PATH}/mautic/media/files:/var/www/html/docroot/media/files:z
  - ${BASE_PATH}/mautic/media/images:/var/www/html/docroot/media/images:z
  - ${BASE_PATH}/cron:/opt/mautic/cron:z
  - mautic-docroot:/var/www/html/docroot:z

services:
  db:
    image: mysql:8.0
    environment:
      - MYSQL_ROOT_PASSWORD=${MYSQL_ROOT_PASSWORD}
      - MYSQL_USER=${MYSQL_USER}
      - MYSQL_PASSWORD=${MYSQL_PASSWORD}
      - MYSQL_DATABASE=${MYSQL_DATABASE}
    ports:
      - "127.0.0.1:3307:3306"
    volumes:
      - ${BASE_PATH}/mysql-data:/var/lib/mysql
      - ./init-strapi-db.sql:/docker-entrypoint-initdb.d/init-strapi-db.sql
    healthcheck:
      test: mysqladmin --user=${MYSQL_USER} --password=${MYSQL_PASSWORD} ping
      start_period: 5s
      interval: 5s
      timeout: 5s
      retries: 10
```

```

networks:
  - default

nginx:
  image: nginx
  volumes:
    - ./nginx.conf:/etc/nginx/conf.d/default.conf
    - mautic-docroot:/var/www/html/docroot:z
    - ./mautic/media/files:/var/www/html/docroot/media/files:z
    - ./mautic/media/images:/var/www/html/docroot/media/images:z
    # TLS material + ACME webroot, shared with the certbot service.
    - certbot-etc:/etc/letsencrypt
    - certbot-www:/var/www/certbot
  depends_on:
    - mautic_web
    - app
    - strapi
  ports:
    - 80:80      # taxnexusapp.com - ACME challenge + 301 to https
    - 443:443     # taxnexusapp.com (app) + cms.taxnexusapp.com (strapi), TLS
    - 8082:8082   # mautic admin (internal/no public TLS in this iteration)
  # Reload every 6h so renewed certs are picked up without a restart.
  command: '/bin/sh -c "while :; do sleep 6h & wait $$(!); nginx -s reload; done & nginx -g
"daemon off;"'
  networks:
    - default

mautic_web:
  image: mautic/mautic:6-fpm
  links:
    - db:mysql
  volumes: *soloai-volumes
  environment:
    - DOCKER_MAUTIC_LOAD_TEST_DATA=${DOCKER_MAUTIC_LOAD_TEST_DATA}
    - DOCKER_MAUTIC_RUN_MIGRATIONS=${DOCKER_MAUTIC_RUN_MIGRATIONS}
  env_file:
    - .mautic_env
  healthcheck:
    test: cgi-fcgi -bind -connect 127.0.0.1:9000
    start_period: 5s
    interval: 5s
    timeout: 5s
    retries: 100
  depends_on:
    db:
      condition: service_healthy
  networks:
    - default

mautic_cron:
  image: mautic/mautic:6-fpm
  links:
    - db:mysql
  volumes: *soloai-volumes
  environment:
    - DOCKER_MAUTIC_ROLE=mautic_cron
  env_file:
    - .mautic_env
  depends_on:
    mautic_web:
      condition: service_healthy
  networks:
    - default

```

```

mautic_worker:
  image: mautic/mautic:6-fpm
  links:
    - db:mysql
  volumes: *soloai-volumes
  environment:
    - DOCKER_MAUTIC_ROLE=mautic_worker
  env_file:
    - .mautic_env
  depends_on:
    mautic_web:
      condition: service_healthy
  networks:
    - default

strapi:
  build:
    context: .
    dockerfile: Dockerfile.strapi
  environment:
    NODE_ENV: development # or production
    DATABASE_CLIENT: mysql
    DATABASE_HOST: ${MYSQL_HOST}
    DATABASE_PORT: ${MYSQL_PORT}
    DATABASE_NAME: strapi_db
    DATABASE_USERNAME: ${MYSQL_STRAPI_USER}
    DATABASE_PASSWORD: ${MYSQL_PASSWORD}
    JWT_SECRET: ${JWT_SECRET}
    ADMIN_JWT_SECRET: ${ADMIN_JWT_SECRET}
    STRAPI_ADMIN_JWT_SECRET: ${ADMIN_JWT_SECRET}
    APP_KEYS: ${APP_KEYS}
    TRANSFER_TOKEN_SALT: ${TRANSFER_TOKEN_SALT}
    API_TOKEN_SALT: ${API_TOKEN_SALT}
    DATABASE_SSL: false
  ports:
    - "127.0.0.1:1337:1337"
  depends_on:
    db:
      condition: service_healthy
  # Removed volume mount - was overwriting built app files
  networks:
    - default

# SvelteKit (TaxNexus) production app - served at https://taxnexusapp.com via nginx.
app:
  build:
    context: .
    dockerfile: Dockerfile.app
  args:
    # Inlined into the client bundle at build - must be the real public origins.
    PUBLIC_STRAPI_URL: https://cms.taxnexusapp.com
    STRAPI_API_URL: https://cms.taxnexusapp.com
    BETTER_AUTH_URL: https://taxnexusapp.com
  env_file:
    - .env
  environment:
    NODE_ENV: production
    # adapter-node runtime
    PORT: 3000
    HOST: 0.0.0.0
    ORIGIN: https://taxnexusapp.com
    # Server-to-Strapi must be HTTPS in production (env validator) - hairpins via nginx.

```

```

    STRAPI_API_URL: https://cms.taxnexusapp.com
    PUBLIC_STRAPI_URL: https://cms.taxnexusapp.com
    BETTER_AUTH_URL: https://taxnexusapp.com
    DATABASE_AUTH_PATH: /app/data/auth.sqlite
volumes:
  # Persist Better Auth SQLite + JSON billing store across deploys.
  - app-data:/app/data
depends_on:
  db:
    condition: service_healthy
networks:
  - default

# TLS issuance + renewal (Let's Encrypt, webroot challenge). Shares volumes with nginx.
certbot:
  image: certbot/certbot
  volumes:
    - certbot-etc:/etc/letsencrypt
    - certbot-www:/var/www/certbot
  # Try renewal twice a day; nginx reloads on its own 6h timer to pick up new certs.
  entrypoint: "/bin/sh -c 'trap exit TERM; while ;; do certbot renew --webroot -w
/var/www/certbot; sleep 12h & wait $$!}; done'"
  networks:
    - default

volumes:
  mysql-data:
    driver: local
    driver_opts:
      type: none
      device: ${BASE_PATH}/mysql-data
      o: bind
  mautic-docroot:
    driver: local
    driver_opts:
      type: none
      device: ${BASE_PATH}/mautic-docroot
      o: bind
  strapi-app:
    driver: local
    driver_opts:
      type: none
      device: ${BASE_PATH}/strapi
      o: bind
  # SvelteKit app persistent state (Better Auth SQLite + JSON billing store).
  # Bind-mounted to a host path (like mysql-data/strapi) so accounts survive
  # container rebuilds AND are not lost to `docker compose down -v` / volume prune.
  # NOTE: the host dir must exist before `up` - `mkdir -p ${BASE_PATH}/app-data`.
  app-data:
    driver: local
    driver_opts:
      type: none
      device: ${BASE_PATH}/app-data
      o: bind
  # Let's Encrypt certs + ACME webroot, shared between nginx and certbot.
  certbot-etc:
    driver: local
  certbot-www:
    driver: local

networks:

```

```
default:
  name: ${COMPOSE_PROJECT_NAME}-docker
```

3. Full Contents — Dockerfile.app

Verbatim from repo root. Multi-stage build: full toolchain builder → slim adapter-node runtime with pruned prod deps (incl. compiled native better-sqlite3).

```
Dockerfile.app repo root

# SvelteKit (TaxNexus) production image – adapter-node standalone server.
# Multi-stage: build with full toolchain, ship a slim runtime with pruned prod deps.

# ---- builder -----
FROM node:20-alpine AS builder

# better-sqlite3 (Better Auth) is a native module – needs a C/C++ toolchain to compile.
RUN apk add --no-cache libc6-compat build-base python3

WORKDIR /app

# Install with a clean, lockfile-exact tree (better caching: deps before source).
COPY package.json package-lock.json ./
RUN npm ci

# Build the app. These ARGs satisfy the env validator (src/lib/server/env.bootstrap),
# which runs at build time because vite sets NODE_ENV=production.
# Only PUBLIC_* values are inlined into the client bundle – pass the REAL public origin there.
# Server-only values (STRAPI_API_URL, DATABASE_URL) are validated for FORMAT only at build
# and overridden at runtime via env_file, so placeholders here are fine and bake no secrets.
ARG PUBLIC_STRAPI_URL=https://cms.taxnexusapp.com
ARG STRAPI_API_URL=https://cms.taxnexusapp.com
ARG DATABASE_URL=mysql://placeholder:placeholder@db:3306/soloai_db
ARG BETTER_AUTH_SECRET=build_time_placeholder_secret_min_32_characters
ARG BETTER_AUTH_URL=https://taxnexusapp.com
ENV NODE_ENV=production \
    PUBLIC_STRAPI_URL=$PUBLIC_STRAPI_URL \
    STRAPI_API_URL=$STRAPI_API_URL \
    DATABASE_URL=$DATABASE_URL \
    BETTER_AUTH_SECRET=$BETTER_AUTH_SECRET \
    BETTER_AUTH_URL=$BETTER_AUTH_URL

COPY . .
RUN npm run build

# Drop dev dependencies; keep the compiled native better-sqlite3 binary.
RUN npm prune --omit=dev

# ---- runner -----
FROM node:20-alpine AS runner

# Runtime needs libc6-compat for the prebuilt/compiled native module.
RUN apk add --no-cache libc6-compat

WORKDIR /app
ENV NODE_ENV=production

# adapter-node server, pruned prod deps (incl. native better-sqlite3), and manifest.
COPY --from=builder /app/build ./build
COPY --from=builder /app/node_modules ./node_modules
```

```

COPY --from=builder /app/package.json ./package.json

# Better Auth SQLite + JSON billing store live here; mounted to a named volume in compose.
RUN mkdir -p /app/data && chown -R node:node /app
USER node

# adapter-node listens on PORT/HOST; ORIGIN is set for CSRF behind the nginx TLS proxy.
ENV PORT=3000 HOST=0.0.0.0
EXPOSE 3000

CMD ["node", "build/index.js"]

```

4. Exact Deployment Sequence

The exact command sequence used, per docs/deploy-runbook.md (Phases 0–7 + ongoing deploys). Run from the repo checkout on the VPS (e.g. /opt/taxnexus).

Phase 1 — VPS prerequisites (one-time)

```

git clone https://github.com/darnell-ux/soloai_starter.git /opt/taxnexus
cd /opt/taxnexus

```

Phase 2 — DNS (do first — TLS depends on it)

Point A records taxnexusapp.com, www, and cms at 2.25.203.69, then verify from the VPS:

```

for h in taxnexusapp.com www.taxnexusapp.com cms.taxnexusapp.com; do
  echo -n "$h -> "; getent hosts "$h" | awk '{print $1}' | head -1 || echo "NO DNS"
done

```

Phase 3 — Secrets + .env

```

cp .env.production.example .env
# Opaque secrets (BETTER_AUTH_SECRET, JWT_SECRET, ADMIN_JWT_SECRET, APP_KEYS, salts):
openssl rand -base64 32
# DB passwords — hex is URL-safe (MYSQL_PASSWORD, MYSQL_ROOT_PASSWORD):
openssl rand -hex 24

```

Phase 4 — Validate before building

```

docker compose config -q      # compose syntax
npm run validate-env          # hard gate — required env present + valid
npm run check:secrets         # no placeholder / low-entropy secrets left

```

Phase 5 — TLS bootstrap (one-time)

```

# 1) Let's Encrypt STAGING dry run (avoid rate limits)
EMAIL=you@yourdomain.com STAGING=1 bash scripts/init-letsencrypt.sh
# 2) real cert (cms auto-included once its DNS resolves)
EMAIL=you@yourdomain.com bash scripts/init-letsencrypt.sh
# ...or force cms in once its A record is confirmed:
EMAIL=you@yourdomain.com INCLUDE_CMS=1 bash scripts/init-letsencrypt.sh

```

Phase 6 — Build + start the stack

```
docker compose build app strapi
docker compose up -d --remove-orphans
docker compose ps
# Equivalent one-shot:
DEPLOY_BRANCH=main bash scripts/deploy.sh
```

Phase 7 — Strapi admin + API token

```
# Create admin at https://cms.taxnexusapp.com/admin, mint a read-only API token,
# put it in .env as STRAPI_API_TOKEN=..., then restart just the app:
docker compose up -d --no-deps app
```

Phase 8 — Verify (see §5 for live output)

```
curl -sSI https://taxnexusapp.com/ | head -20
curl -sSI https://cms.taxnexusapp.com/admin | head -5
docker compose logs --tail=50 app strapi nginx
```

Ongoing deploys & rollback

```
# Automated: push to main -> .github/workflows/deploy.yml SSHes in and runs scripts/deploy.sh
# Manual on the VPS:
DEPLOY_BRANCH=main bash scripts/deploy.sh

# Rollback to a previous good commit (app-data + mysql-data volumes persist):
git reset --hard <previous-good-sha>
docker compose build app strapi && docker compose up -d
```

5. Verification — Live Output

Captured live against production on 2026-08-14. Raw, unedited output.

5.1 curl -I https://taxnexusapp.com HTTP/2 200

```
HTTP/2 200
server: nginx/1.31.1
date: Fri, 14 Aug 2026 02:18:44 GMT
content-type: text/html
content-length: 36437
vary: Accept-Encoding
cache-control: private, max-age=60
etag: "bitsup"
link: <./_app/immutable/assets/0.e1hW4So6.css>; rel="preload"; as="style"; nopush,
<./_app/immutable/entry/start.D9DR3DwG.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/Bc57U4vP.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/BKCYxWom.js>; rel="modulepreload"; nopush, <./_app/immutable/entry/app.q1FM-
7tX.js>; rel="modulepreload"; nopush, <./_app/immutable/chunks/B04Ne0Gr.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/BZbeTG8o.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/CXsyGL_h.js>; rel="modulepreload"; nopush,
<./_app/immutable/nodes/0.Cn0AtPvI.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/Ba10TM78.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/DqIMTtdv.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/jZTgUXcX.js>; rel="modulepreload"; nopush,
```



```

<./_app/immutable/chunks/CEl2_IX5.js>; rel="modulepreload"; nopush,
<./_app/immutable/nodes/2.DPwbJLS6.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/CssKZECq.js>; rel="modulepreload"; nopush,
<./_app/immutable/chunks/DdFwKHpx.js>; rel="modulepreload"; nopush
x-sveltekit-page: true
strict-transport-security: max-age=31536000; includeSubDomains

```

App is up: **HTTP/2 200** from nginx, SvelteKit page (x-sveltekit-page: true), HSTS enabled.

5.2 curl -I https://cms.taxnexusapp.com HTTP/2 302

```

HTTP/2 302
server: nginx/1.31.1
date: Fri, 14 Aug 2026 02:18:44 GMT
content-type: text/html; charset=utf-8
content-length: 43
content-security-policy: connect-src 'self' https;;img-src 'self' data: blob: https://market-
assets.strapi.io;media-src 'self' data: blob;;default-src 'self';base-uri 'self';font-src 'self' https:
data;;form-action 'self';frame-ancestors 'self';object-src 'none';script-src 'self';script-src-attr
'none';style-src 'self' https: 'unsafe-inline'
referrer-policy: no-referrer
strict-transport-security: max-age=31536000; includeSubDomains
x-content-type-options: nosniff
x-dns-prefetch-control: off
x-download-options: noopen
x-frame-options: SAMEORIGIN
x-permitted-cross-domain-policies: none
vary: Origin
access-control-allow-origin:
access-control-allow-credentials: true
location: http://localhost:1337/admin
x-powered-by: Strapi <strapi.io>
strict-transport-security: max-age=31536000; includeSubDomains

```

Strapi answers over valid TLS and redirects / → /admin (x-powered-by: Strapi) — expected healthy behavior for the CMS root.

5.3 openssl s_client -connect taxnexusapp.com:443 -servername taxnexusapp.com

```

depth=4 C = US, 0 = Internet Security Research Group, CN = ISRG Root X1
verify return:1
depth=3 C = US, 0 = Internet Security Research Group, CN = ISRG Root X2
verify return:1
depth=2 C = US, 0 = ISRG, CN = Root YE
verify return:1
depth=1 C = US, 0 = Let's Encrypt, CN = YE2
verify return:1
depth=0 CN = taxnexusapp.com
verify return:1
CONNECTED(00000005)
write W BLOCK
---
Certificate chain
0 s:/CN=taxnexusapp.com
i:/C=US/0=Let's Encrypt/CN=YE2
1 s:/C=US/0=Let's Encrypt/CN=YE2
i:/C=US/0=ISRG/CN=Root YE
2 s:/C=US/0=ISRG/CN=Root YE

```

```

i:/C=US/O=Internet Security Research Group/CN=ISRG Root X2
3 s:/C=US/O=Internet Security Research Group/CN=ISRG Root X2
i:/C=US/O=Internet Security Research Group/CN=ISRG Root X1
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIDtjCCAzygAwIBAgISBV34EpTsk+2Y0uR0KBya1LCbMAoGCCqGSM49BAMDMx
CzAJBgNVBAYTAlVTMRYwFAYDVQQKEw1MZXQncyBFbmNyeXB0MQwwCgYDVQQDEwNZ
RTIwHhcNMjYwNzA0MjEwNDU0WhcNMjYxMDAyMjEwNDUzWjAaMRgwFgYDVQQDEw90
YXhuZXh1c2FwcC5jb20wWTATBgqhkhj0PQIBBgqhkhj0PQMBBwNCAAQwTwy7sXC3
QVIedYNNpyz06uPSegsISUhlSNBN7q9uo5Q5jo0gDXBnysCRAW4dVbHTQ2jbfYy
lcDP5+BYK3aoo4ICRzCCAKMwDgYDVR0PAQH/BAQDAgeAMBMGA1UdJQQMMAoGCCsG
AQUFBwMBMAwGA1UdEwEB/wQCMAAwHQYDVR00BBYEFNsRgWsZeVQUBYhXlnUiMCR9
LhgpMB8GA1UdIwQYMBaAFLlZ8o7PIvCG0zdI/3YUGLqC2FWMHMDMGCCsGAQUFBwEB
BCcwJTAjBggrBgEFBQcwAoYXaHR0cDovL3llMi5pLmxlbmNyLm9yZyZwRAYDVR0R
BD0w04ITY21zLnRheG5leHVzYXBwLmNvbYIPdGF4bmV4dXNhchAuY29tghN3d3cu
dGF4bmV4dXNhchAuY29tMBMGA1UdIAQMMAowCAYGZ4EMAQIBMC4GA1UdHwQnMCUw
I6AhoB+GHWh0dHA6L9y5ZTIuYy5sZW5jci5vcmcvMjguY3J3SMTIBDAYKwYBBAHW
eQIEAgSB/QSB+gD4AHUAwjF+V0Uzo0XufzjespBB68fCIVoiv3/Vta12mtk0Us0A
AAGfLyhmZgAABAMARjBEAiBvzQb2IuY9AauX6XYQK6Q9tXhSda0KM2TI+D5cbZZW
BAIgR0iq2s2VSSDEL+x7a5+f0o3/2ea/vV30cxoTKS4RmHsAfwAai51rD/6/gbR5
0cbSMQqG1tEC1PBG4hgsneNfXiYl7wAAAZ8vKGqeAagAAAUAJjY0LQQDAEgwRgIh
ALLZEJzU4fNk0rSMMeBX4Eq9GIgT1SF9gso05ZDVk8g6AiEA3wwViTQLG980ZUK0
aEcGcyWBxmSaKsSPI3JDBCK6f4IwCgYIKoZIzj0EAwMDAAAwZQIwKSLC4xQBWHKT
XTEhSkoYJuj7MqPnCMT4Z0MVhRsK34pDTTrFrwSJGsEVJ3RyJggHYAjEAmkRdJL4T
xskPxZ5+EMzYs/jft/AdGBn1qxUluVK8xfyAuPwKj+0UZn5iHqj6dtY
-----END CERTIFICATE-----
subject=/CN=taxnexusapp.com
issuer=/C=US/O=Let's Encrypt/CN=YE2
---
No client certificate CA names sent
Server Temp Key: ECDH, X25519, 253 bits
---
SSL handshake has read 3828 bytes and written 375 bytes
---
New, TLSv1/SSLv3, Cipher is AEAD-CHACHA20-POLY1305-SHA256
Server public key is 256 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
SSL-Session:
    Protocol : TLSv1.3
    Cipher : AEAD-CHACHA20-POLY1305-SHA256
    Session-ID:
    Session-ID-ctx:
    Master-Key:
    Start Time: 1786673924
    Timeout : 7200 (sec)
    Verify return code: 0 (ok)
---
poll errorFAILED

```

Chain verifies to ISRG Root (verify return:1 at every depth); issuer **Let's Encrypt**; the SAN block covers cms.taxnexusapp.com, taxnexusapp.com, and www.taxnexusapp.com — proving the CMS hairpin cert requirement from §1 is satisfied.

6. Troubleshooting — Issues Encountered & Resolved

Real problems hit during the Week 3 deploy and how each was fixed.

Issue	Cause / Symptom	Resolution
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Compose volume anchor typo	A malformed YAML anchor for the shared Mautic volumes (<code>Trx-soloai-volumes</code> - style typo) broke the <code>&soloai-volumes</code> / <code>*soloai-volumes</code> alias, so <code>docker compose config</code> failed to parse.	Corrected the anchor to <code>x-soloai-volumes: &soloai-volumes</code> and referenced it as <code>volumes: *soloai-volumes</code> on the mautic services. <code>docker compose config -q</code> now passes.
MySQL exposed on <code>0.0.0.0:3307</code>	The <code>db</code> port mapping bound MySQL to all interfaces (<code>3307:3306</code>), exposing the database to the public internet on the VPS IP.	Bound the published port to loopback only: <code>"127.0.0.1:3307:3306"</code> . MySQL is now reachable only from the host/containers, not externally.
Strapi React peer deps missing	The Strapi admin build failed because required React peer dependencies were not installed in <code>Dockerfile.strapi</code> .	Added the missing React peer deps to the Strapi image install step so the admin panel builds cleanly during <code>docker compose build strapi</code> .
cms dropped from cert until DNS ready	<code>cms.taxnexusapp.com</code> 's A record wasn't propagated when TLS first ran, so Let's Encrypt issued apex+www only; the app's server-side HTTPS calls to the CMS failed until it was added.	Waited for the cms A record to resolve, then re-issued with <code>EMAIL=...</code> <code>INCLUDE_CMS=1 bash scripts/init-letsencrypt.sh</code> . SAN now includes cms (verified in §5.3).
SSH key-only auth (Hostinger KVM2)	Password authentication is disabled on the KVM2; deploys/SSH fail with "Permission denied (publickey)" if the key isn't loaded.	Standardized on SSH key auth — the deploy key is registered on the VPS and stored as the GitHub Actions deploy secret; no password login path is used.
Git branch divergence (VPS vs origin)	The VPS checkout had drifted from <code>origin/main</code> , so <code>git pull</code> reported divergent branches and deploys picked up stale code.	Made deploys authoritative to origin — <code>scripts/deploy.sh</code> does a <code>git reset --hard origin/main</code> before rebuilding, eliminating drift on every deploy.

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